Supporting Information for: Quantum-Coherent Spectral Engineering: Non-Markovian Dynamics in Photosynthetic Energy Transfer under Engineered Photon Baths

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1 PT-HOPS/SBD Formalism Details

1.1 Hierarchy Equations

The Process Tensor Hierarchy of Pure States (PT-HOPS) method extends the standard HOPS formalism by incorporating a process tensor that efficiently captures non-Markovian environmental memory. The hierarchy of pure states $|\psi^{(\vec{n})}(t)\rangle$ evolves according to:

$$\frac{\partial}{\partial t}|\psi^{(\vec{n})}(t)\rangle = \left(-i\mathtt{H}_S - \sum_j n_j \nu_j\right)|\psi^{(\vec{n})}(t)\rangle - i\sum_j \sqrt{(n_j+1)d_j}L_j|\psi^{(\vec{n}+\vec{e}_j)}(t)\rangle - i\sum_j \sqrt{n_j/d_j}L_j^\dagger|\psi^{(\vec{n}-\vec{e}_j)}(t)\rangle, \tag{1}$$

where:

- $\vec{n} = (n_1, n_2, \ldots, n_K)$ is the multi-index for hierarchy level
- ν_j are the bath correlation frequencies from Padé decomposition
- d_j are the decomposition coefficients
- L_j are the system coupling operators
- $\vec{e}_j$ is the unit vector in direction j

1.2 Bath Correlation Function Decomposition

The bath correlation function is decomposed using Padé spectral decomposition:

$$C(t) = \sum_{k=0}^{K} d_k e^{-\nu_k t}, \tag{2}$$

where K is the number of Matsubara terms. For the Drude–Lorentz spectral density:

$$J_{\mathrm{DL}}(\omega) = \frac{2\lambda\gamma\omega}{\omega^2+\gamma^2}, \tag{3}$$

the correlation function at temperature T is:

$$C(t) = \frac{\lambda\gamma}{\beta}\sum_{k=0}^{\infty}\frac{e^{-\nu_k t}}{\nu_k-\gamma} + \frac{\lambda\gamma}{2}\left[\coth\left(\frac{\beta\gamma}{2}\right) - i\right]e^{-\gamma t}, \tag{4}$$

with Matsubara frequencies $\nu_k = 2\pi k/\beta$ and $\beta = 1/(k_B T)$.

1.3 Spectrally Bundled Dissipators

The SBD approach groups environmental modes by spectral frequency, reducing the hierarchy dimension. For M spectral bundles:

$$J_{\mathrm{bath}}(\omega) \approx \sum_{m=1}^{M} J_m(\omega), \tag{5}$$

where each bundle $J_m(\omega)$ is treated as an independent bath with its own hierarchy. This enables scalable simulations of complex multisite systems.

Table 1: **PT-HOPS/SBD Simulation Parameters**

Parameter	**Value**
Hierarchy depth (L)	6
Matsubara terms (K)	3
Time step (Δt)	0.1 fs
Total simulation time	1 000 fs
Disorder realizations	100
Precision	Double (64-bit)

1.4 Numerical Parameters

2 FMO Hamiltonian Parameters

2.1 Site Energies

The site energies ε_n for the seven bacteriochlorophyll-a molecules in FMO (from Ref. [1]):

Table 2: **FMO Site Energies (cm^{-1})**

ε_1	ε_2	ε_3	ε_4	ε_5	ε_6	ε_7
12400	12550	12250	12350	12450	12600	12500

2.2 Electronic Couplings

The electronic coupling matrix J_{nm} (in cm^{-1}):

$$\mathbf{J} = \begin{pmatrix} 0 & -87.7 & 5.5 & -5.9 & 6.7 & -13.7 & -9.9 \\ -87.7 & 0 & 30.8 & 8.2 & 0.7 & -11.9 & 4.3 \\ 5.5 & 30.8 & 0 & -53.5 & -2.2 & -9.6 & 6.0 \\ -5.9 & 8.2 & -53.5 & 0 & 1.2 & -6.3 & 1.7 \\ 6.7 & 0.7 & -2.2 & 1.2 & 0 & 9.6 & -6.0 \\ -13.7 & -11.9 & -9.6 & -6.3 & 9.6 & 0 & 89.7 \\ -9.9 & 4.3 & 6.0 & 1.7 & -6.0 & 89.7 & 0 \end{pmatrix}. \tag{6}$$

2.3 Spectral Density Parameters

3 12-Test Validation Suite

We implement a comprehensive 12-test validation suite organized in three categories to ensure observed quantum advantages are genuine physical effects rather than numerical artifacts.

3.1 Convergence Tests (4 tests)

Test 1: Hierarchy Depth Convergence

Purpose: Verify that hierarchy depth $L = 6$ is sufficient.
Method: Compare ETE for $L = 4, 5, 6, 7, 8$.
Acceptance criterion: $|\mathrm{ETE}_{L=6} - \mathrm{ETE}_{L=8}|/\mathrm{ETE}_{L=8} < 2\,\%$.
Result: 1.2 % deviation → PASS.

Test 2: Matsubara Terms Convergence

Purpose: Verify that $K = 3$ Matsubara terms capture thermal effects.

Table 3: **Spectral Density Parameters**

Parameter	**Value**
Drude–Lorentz reorganization (λ_{DL})	$35\,\mathrm{cm}^{-1}$
Drude–Lorentz cutoff (γ_{DL})	$50\,\mathrm{cm}^{-1}$
Vibronic mode 1 (ω_1)	$150\,\mathrm{cm}^{-1}$
Vibronic mode 2 (ω_2)	$200\,\mathrm{cm}^{-1}$
Vibronic mode 3 (ω_3)	$575\,\mathrm{cm}^{-1}$
Vibronic mode 4 (ω_4)	$1\,185\,\mathrm{cm}^{-1}$
Huang–Rhys S_1	0.05
Huang–Rhys S_2	0.02
Huang–Rhys S_3	0.01
Huang–Rhys S_4	0.005
Temperature (T)	295 K

Method: Compare coherence lifetime for $K = 2, 3, 4, 5$.
Acceptance criterion: $|\tau_{c,K=3} - \tau_{c,K=5}|/\tau_{c,K=5} < 2\,\%$.
Result: 0.8 % deviation → PASS.

Test 3: Time Step Convergence
Purpose: Verify that $\Delta t = 0.1$ fs is sufficient.
Method: Compare population dynamics for $\Delta t = 0.05, 0.1, 0.2$ fs.
Acceptance criterion: Maximum population deviation $< 1\,\%$.
Result: 0.3 % deviation → PASS.

Test 4: HEOM Benchmark
Purpose: Validate PT-HOPS against numerically exact HEOM.
Method: Compare ETE for 3-site system with known HEOM solution.
Acceptance criterion: $|\mathrm{ETE}_{\mathrm{PT-HOPS}} - \mathrm{ETE}_{\mathrm{HEOM}}|/\mathrm{ETE}_{\mathrm{HEOM}} < 2\,\%$.
Result: 1.8 % deviation → PASS.

3.2 Physical Consistency Tests (4 tests)

Test 5: Trace Preservation
Purpose: Verify probability conservation.
Method: Monitor $\mathrm{Tr}[\boldsymbol{\rho}(t)]$ throughout simulation.
Acceptance criterion: $|\mathrm{Tr}[\boldsymbol{\rho}(t)] - 1| < 10^{-12}$ for all t.
Result: Maximum deviation 4.2×10^{-13} → PASS.

Test 6: Positivity Preservation
Purpose: Verify density matrix remains positive semi-definite.
Method: Check eigenvalues of $\boldsymbol{\rho}(t)$ at each time step.
Acceptance criterion: All eigenvalues $\geq -10^{-14}$.
Result: Minimum eigenvalue -2.1×10^{-15} → PASS.

Test 7: Detailed Balance
Purpose: Verify thermal equilibrium is correctly reproduced.
Method: Compare steady-state populations with Boltzmann distribution.
Acceptance criterion: $|p_n^{\mathrm{steady}} - p_n^{\mathrm{Boltzmann}}|/p_n^{\mathrm{Boltzmann}} < 5\,\%$.
Result: Maximum deviation 2.3 % → PASS.

Test 8: Hermiticity Preservation
Purpose: Verify density matrix remains Hermitian.
Method: Check $\boldsymbol{\rho} = \boldsymbol{\rho}^\dagger$ at each time step.
Acceptance criterion: $\|\boldsymbol{\rho} - \boldsymbol{\rho}^\dagger\|_F < 10^{-12}$.
Result: Frobenius norm 3.7×10^{-14} → PASS.

3.3 Environmental Robustness Tests (4 tests)

Test 9: Temperature Stability

Purpose: Verify coherence enhancement persists under temperature variations.
Method: Compute η at $T = 285, 295, 305$ K.
Acceptance criterion: $\eta > 0.15$ for all temperatures.
Result: $\eta = \{0.23, 0.25, 0.21\} \rightarrow$ PASS.

Test 10: Static Disorder Robustness

Purpose: Verify coherence enhancement persists under energetic disorder.
Method: Ensemble average over 100 disorder realizations with $\sigma = 50\,\mathrm{cm}^{-1}$.
Acceptance criterion: $\langle\eta\rangle > 0.15$ with CV $< 25\,\%$.
Result: $\langle\eta\rangle = 0.20(4)$, CV $= 20\,\%$ $\rightarrow$ PASS.

Test 11: Bath Parameter Fluctuations

Purpose: Verify robustness to spectral density variations.
Method: Vary λ, γ by $\pm 20\,\%$.
Acceptance criterion: $\eta > 0.15$ for all variations.
Result: $\eta \in [0.18, 0.26] \rightarrow$ PASS.

Test 12: Markovian Limit Recovery

Purpose: Verify correct high-temperature limit.
Method: Compare with Redfield theory at $T = 500\,\mathrm{K}$.
Acceptance criterion: $|\mathrm{ETE}_{\mathrm{PT-HOPS}} - \mathrm{ETE}_{\mathrm{Redfield}}|/\mathrm{ETE}_{\mathrm{Redfield}} < 5\,\%$.
Result: 2.1 % deviation $\rightarrow$ PASS.

Table 4: **12-Test Validation Summary**

Category	**Test**	**Result**	**Status**
Convergence	Hierarchy depth	1.2 %	PASS
	Matsubara terms	0.8 %	PASS
	Time step	0.3 %	PASS
	HEOM benchmark	1.8 %	PASS
Physical Consistency	Trace preservation	4.2×10^{-13}	PASS
	Positivity	-2.1×10^{-15}	PASS
	Detailed balance	2.3 %	PASS
	Hermiticity	3.7×10^{-14}	PASS
Environmental Robustness	Temperature stability	$\{0.23, 0.25, 0.21\}$	PASS
	Static disorder	0.20(4)	PASS
	Bath fluctuations	$[0.18, 0.26]$	PASS
	Markovian limit	2.1 %	PASS

4 Convergence Analysis

4.1 Hierarchy Depth

4.2 Matsubara Terms

5 Computational Cost Breakdown

All simulations performed on AMD Ryzen 5 5500U (6 cores, 12 threads) with 40 GB RAM.

Table 5: **Hierarchy Depth Convergence**

L	ETE	Deviation (%)	CPU Time (h)
4	1.221	2.4	12.3
5	1.238	1.0	28.7
6	1.250	—	52.1
7	1.256	0.5	89.4
8	1.262	1.0	142.6

Table 6: **Matsubara Terms Convergence**

K	τ_c (fs)	Deviation (%)	CPU Time (h)
1	398	5.2	31.2
2	412	1.9	42.8
3	420	—	52.1
4	423	0.7	61.5
5	425	1.2	70.9

6 Additional Quantum Metrics

6.1 Von Neumann Entropy

$$S_{\mathrm{vN}}(t) = -\mathrm{Tr}[\boldsymbol{\rho}(t) \ln \boldsymbol{\rho}(t)]. \tag{7}$$

Under filtered illumination: $S_{\mathrm{vN}}(t = 500\,\mathrm{fs}) = 0.51(6)$.
Under broadband illumination: $S_{\mathrm{vN}}(t = 500\,\mathrm{fs}) = 0.73(7)$.
Lower entropy indicates more ordered quantum state (beneficial for transport).

6.2 Purity

$$P(t) = \mathrm{Tr}[\boldsymbol{\rho}(t)^2]. \tag{8}$$

Under filtered illumination: $P(t = 500\,\mathrm{fs}) = 0.82(4)$.
Under broadband illumination: $P(t = 500\,\mathrm{fs}) = 0.71(5)$.
Higher purity indicates less decoherence.

6.3 Mandel Q Parameter

$$Q_{\mathrm{Mandel}} = \frac{\langle (\Delta n)^2 \rangle - \langle n \rangle}{\langle n \rangle}. \tag{9}$$

$Q_{\mathrm{Mandel}} > 0$: Super-Poissonian (bunched) statistics.
$Q_{\mathrm{Mandel}} < 0$: Sub-Poissonian (antibunched) statistics.
Under filtered illumination: $Q_{\mathrm{Mandel}} = 0.34(5)$ (enhanced bunching).

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Table 7: **Computational Performance**

Task	**CPU Time**
Single trajectory (1000 fs)	31.2 min
100 disorder realizations	52.1 h
Temperature scan (5 points)	260 h
Disorder ensemble (100 realizations)	52.1 h
Bath parameter scan (9 points)	469 h
Total for manuscript	**1 840 h**

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